NATIONAL JUNIOR COLLEGE
2008 JC 2/ IP 4 PRELIMINARY EXAMINATIONS

MATHEMATICS **8863**
Higher 1

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Sheet

3 hours

28 Aug 2008 Thursday 0905 – 1205 hrs

INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group, on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **ALL** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

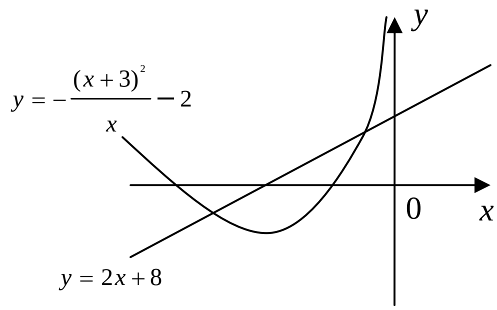
The number of marks is given in the brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This question paper consists of 8 printed pages

Section A: Pure Mathematics [40 Marks]

1 Variables x and y are related by the equation $y = \frac{1}{512}\left(5 - \frac{x}{6}\right)^8$. Given that both x and y vary with time, find the value of x when the rate of change of x is 3 times the rate of change of y. **[4]**

2 The diagram below shows the curve $y = -\frac{(x+3)^2}{x} - 2$ and the line $y = 2x + 8$.



Find $\int -\frac{(x+3)^2}{x} - 2 \, dx$ and hence or otherwise determine the area of the region between the curve and the line $y = 2x + 8$. **[4]**

3 The functions f, g and h are defined for $x \in \mathbb{R}$ by

$$\mathrm{f} : x \mapsto x^3 + x, \ x > 0,$$
$$\mathrm{g} : x \mapsto 3\ln(x - 2c), \ x > 2c,$$
$$\mathrm{h} : x \mapsto \frac{x^2}{5} + 1, \ x > 0.$$

(i) Define, in a similar form, the function h^{-1} and state its range. **[3]**

(ii) Given that $\mathrm{gh}(5) = 6$, find the value of c. **[3]**

4 Sketch the graphs of $y = |2x - 3|$ and $y = 2x^2 + 7x - 2$ on a single diagram. **[1]**

Hence solve the following inequalities exactly:

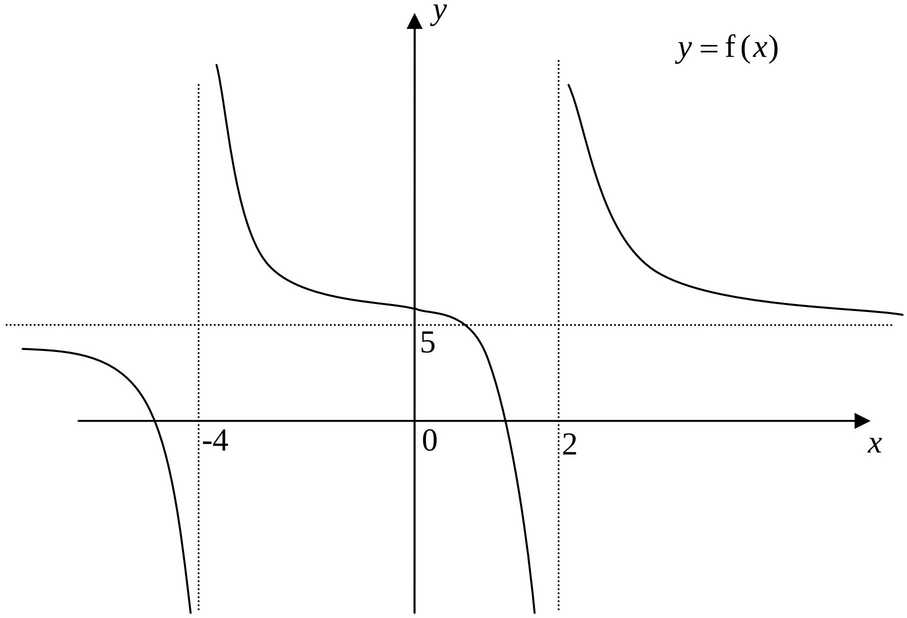
(i) $\left|x - \frac{3}{2}\right| + 1 < x^2 + \frac{7}{2}x$ **[3]**

(ii) $\left|2e^x - 3\right| < 2e^{2x} + 7e^x - 2$. **[2]**

5 Given that $4x^2-4x+5=\left(Ax+B\right)^2+C$ for all values of x, find the constants A, B and C. Hence or otherwise, find

$$\int \frac{4x^2-4x+5}{\left(1-2x\right)^3}\,\mathrm{d}x.$$ **[6]**

6



(i) The diagram above shows a sketch of the curve $y=\mathrm{f}(x)$.

Sketch the graph of $y=\mathrm{f}\,'(x)$, showing clearly the asymptotes. **[4]**

(ii) Given that $\mathrm{f}\left(x\right)=p+\dfrac{3}{x+q}+\dfrac{8}{x+r}$,

where *p*, *q* and *r* are constants, and $q<r$.

The equations of the asymptotes, shown in the diagram, are $x=-4$, $x=2$ and $y=5$.

State the values of *p*, *q* and *r*. **[2]**

7 **(a)** Given that

$$5\left(2^x\right)-2^{-x}=1,$$

find the exact value of x. **[4]**

(b) Given that $\log_x t=3.5$ and $\log_x y=2.7$, find

(i) $\log_x ty^2$ **[2]**

(ii) $\log_y t$. **[2]**

Section B: Statistics [60 Marks]

8 The shop, Tasty Juice, is well known for its watermelon juice and durian milkshake. The mass of a cup of watermelon juice has a mean of 125g and a standard deviation of 5g while the mass of a cup of durian milkshake has a mean of 150g and a standard deviation of 6g.

(i) Calculate the probability that the total mass of 45 cups of watermelon juice will exceed 5.6kg. **[3]**

(ii) 58 cups of durian milkshake are selected at random. Find the probability that their mean mass is at most 148g. **[2]**

9 A group of JC1 Project Work students conducted a price check on hawker stalls selling chicken rice in Singapore. From the 2500 stalls in all the districts of Singapore, the group numbered these stalls and arranged them in numerical order. After which, the 10^{th}, 20^{th}, 30^{th} stall and so on were picked.

(i) Name the type of sampling method that is best described above. **[1]**

(ii) Explain why this method may be inappropriate. **[1]**

(iii) Suggest a better sampling method and describe how it can be carried out. **[3]**

10 In a highly competitive organization, employees are graded according to their performance levels annually as shown in the following probability distribution:

Performance Level	Excellent	Good	Average	Poor
Probability	0.04	0.26	0.43	0.27

The performance of each employee is independent.

(i) Find the greatest value of n if, in a random sample of n employees, the probability of at least 1 employee with good performance is at most 0.96. **[3]**

(ii) In a particular year, there are 200 employees in the organization. Using a suitable approximation, find the probability that there are more than 7 employees with excellent performance. **[3]**

(iii) In a division consisting of 20 employees, find the probability that there are at least 12 employees with average or poor performance, given that there are less than 17 employees with average or poor performance. **[3]**

11 Yau Yan Slimming Centre offers a particular slimming package for ladies. Based on past records, the mean weight of a lady was 50 kg after treatment.

In January 2008, a sample of 50 customers was randomly chosen and the weight, x kg, of each customer after treatment was recorded. The data were summarised by

$$\sum(x-50)=54 \text{ and } \sum(x-50)^2=1226.$$

(i) Find the unbiased estimate of the population variance. **[1]**

(ii) The centre manager claimed that the mean weight after treatment had changed. Test at 10% level of significance, the branch manager's claim. Explain, in context to the question, the meaning of "10% level of significance". **[4]**

In June 2008, the centre manager found that the standard deviation of the weight of a customer after treatment was 6 kg.

(iii) She analysed another random sample of 20 and when a test was conducted at 5% significance level, there was sufficient evidence to justify that the mean weight after treatment had reduced. Find an estimate of the maximum total weight from this sample. Give your answer to the nearest integer. **[3]**

(iv) State any assumption(s) necessary for the validity of the test. **[1]**

12(a) Two events A and B are independent and defined in the same finite sample space. Given that $\mathrm{P}(A)=\dfrac{1}{4}$ and $\mathrm{P}(A\cup B)=\dfrac{2}{3}$, find

(i) $\mathrm{P}(B)$ **[2]**

(ii) $\mathrm{P}(B|A')$ **[1]**

(iii) $\mathrm{P}(A\cap B')$. **[2]**

(b) [In this question, give all your answers to 4 decimal places.]
During the first day of a Junior College enrichment week, a student chooses either swimming or cooking as his morning activity and either table tennis, badminton or shooting as his afternoon activity. Past records showed that for morning activities, 65% of the students chose swimming and for afternoon activities, 55% chose table tennis and 20% chose badminton.
The morning choices are independent of the afternoon choices.
Find the probability that a student selected at random chooses

(i) swimming in the morning and table tennis in the afternoon. **[1]**

(ii) neither swimming nor shooting. **[1]**

Of 3 randomly selected students, what is the probability that two of them choose swimming in the morning and table tennis in the afternoon while the other one chooses neither swimming nor shooting. **[2]**

13(a) Song Car Rental has a fleet of large number of cars. These cars have to be serviced at the end of each year. The costs of servicing a randomly chosen car at two available workshops A and B are normally distributed with means and standard deviations as follow:

	Mean	**Standard Deviation**
Workshop A	\$240	\$30
Workshop B	\$265	\$10

(i) The owner of the car rental company decided that he would send his cars to workshop A if there is a probability of at least 0.8 that the cost of servicing a randomly chosen car by workshop A does not exceed \$$\alpha$. Find this least integer value of α. **[2]**

(ii) Five randomly chosen cars from the fleet are to be serviced by workshop A and another five randomly chosen cars by workshop B. Find the probability that the difference in the cost of servicing is at least \$200. **[4]**

(b) Dioyota and Tonda cars are imported in by several parallel importers. The prices of Dioyota and Tonda cars are normally distributed and are reflected in the table below.

	Mean (in thousands of dollars)	**Standard Deviation (in thousands of dollars)**
Dioyota	\$68	\$2.50
Tonda	\$73	\$3

Find the probability that the price of 3 randomly chosen Dioyota cars will exceed 2.5 times the price of a Tonda car. **[4]**

14 After a major renovation at the computer-mart Sim Lum Square, the manager Keith carried out a study to investigate the relationship between the renovation cost of the shops and their sales figures in the first six months after re-opening. The data below were collected from eight randomly chosen shops, in thousands of dollars.

Shop No.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Renovation cost, *X* (in thousands of dollars)	19.5	23	28.1	31.6	35	38.1	46.5	49.2
Sales figure, *Y* (in thousands of dollars)	55.5	a	77.2	78.7	79	88	73.4	94.9

(i) Keith found that the least square regression line of the above data is $y = 44.209 + bx$. It was known that this line and the regression line of x on y both pass through the point (33.9, 76.2). Find the value of b to 5 significant figures and show that a is 62.9. **[3]**

(ii) Calculate the correlation coefficient, and find the line of regression of x on y. **[3]**

(iii) Give a sketch of the scatter diagram for the data. Identify a data pair, giving a reason, which should be regarded as a suspect. **[3]**

(iv) Explain if it is appropriate to use the given regression line in (i) to predict the renovation cost when given the sales figures. **[1]**

(v) Hence, use an appropriate regression line to find the renovation cost when the sales figure is \$80000. Comment on the validity of this prediction. **[3]**

End of Paper

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INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group and subject tutor's name in the spaces provided on the cover sheet and attached it on top of your answer paper.

Circle the questions you have attempted and arrange your answers in **NUMERICAL ORDER**.

Name:____________________________ Registration Number:______________

Subject Tutorial Group:_____________ Subject Tutor:____________________

Section A	Marks	Max. Marks	Question	Marks	Max. Marks
1		**4**	8		**5**
2		**4**	9		**5**
3		**6**	10		**9**
4		**6**	11		**9**
5		**6**	12		**9**
6		**6**	13		**10**
7		**8**	14		**13**
Subtotal		**40**	**Subtotal**		**60**
Total		**100**	**Grade**		